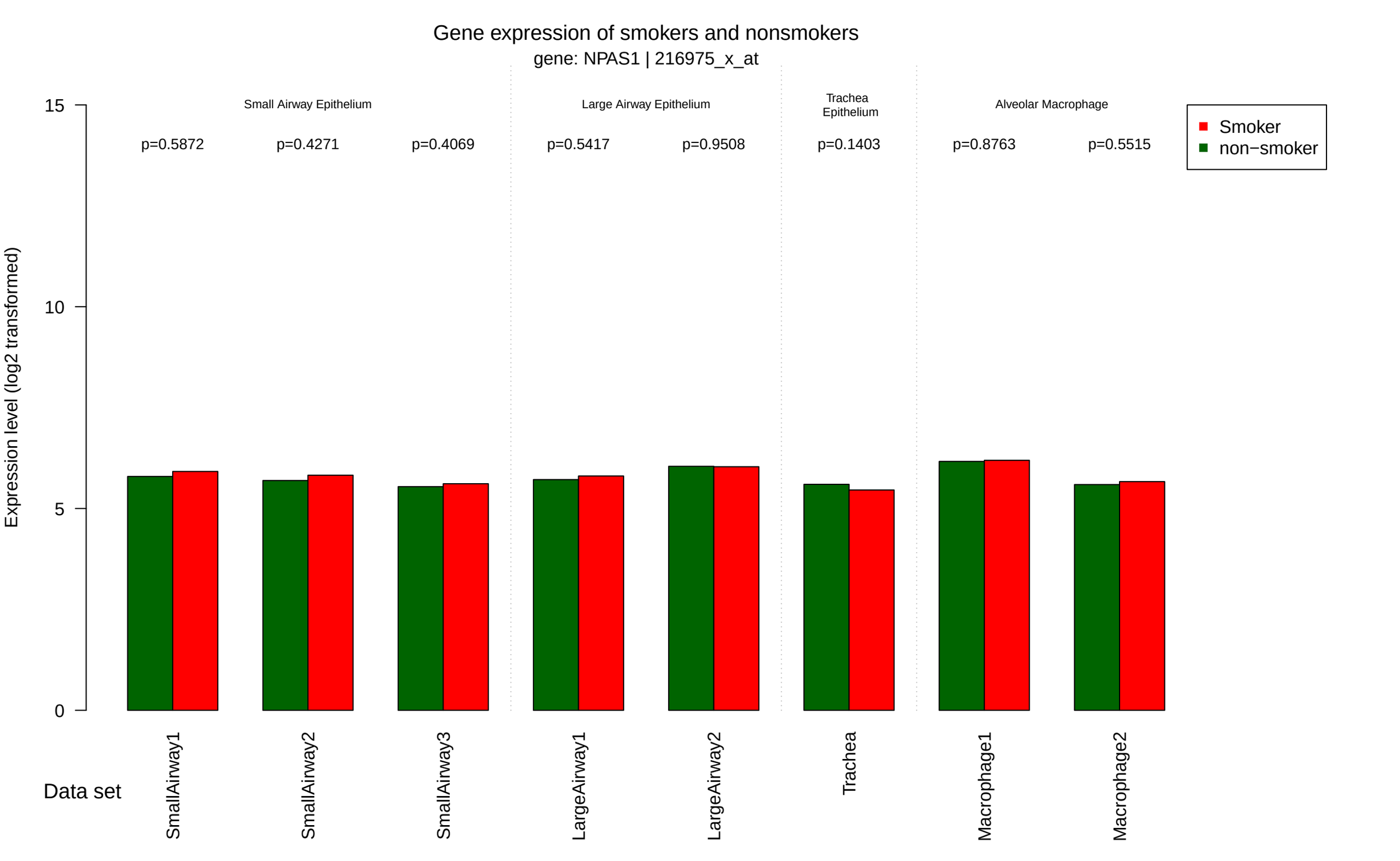
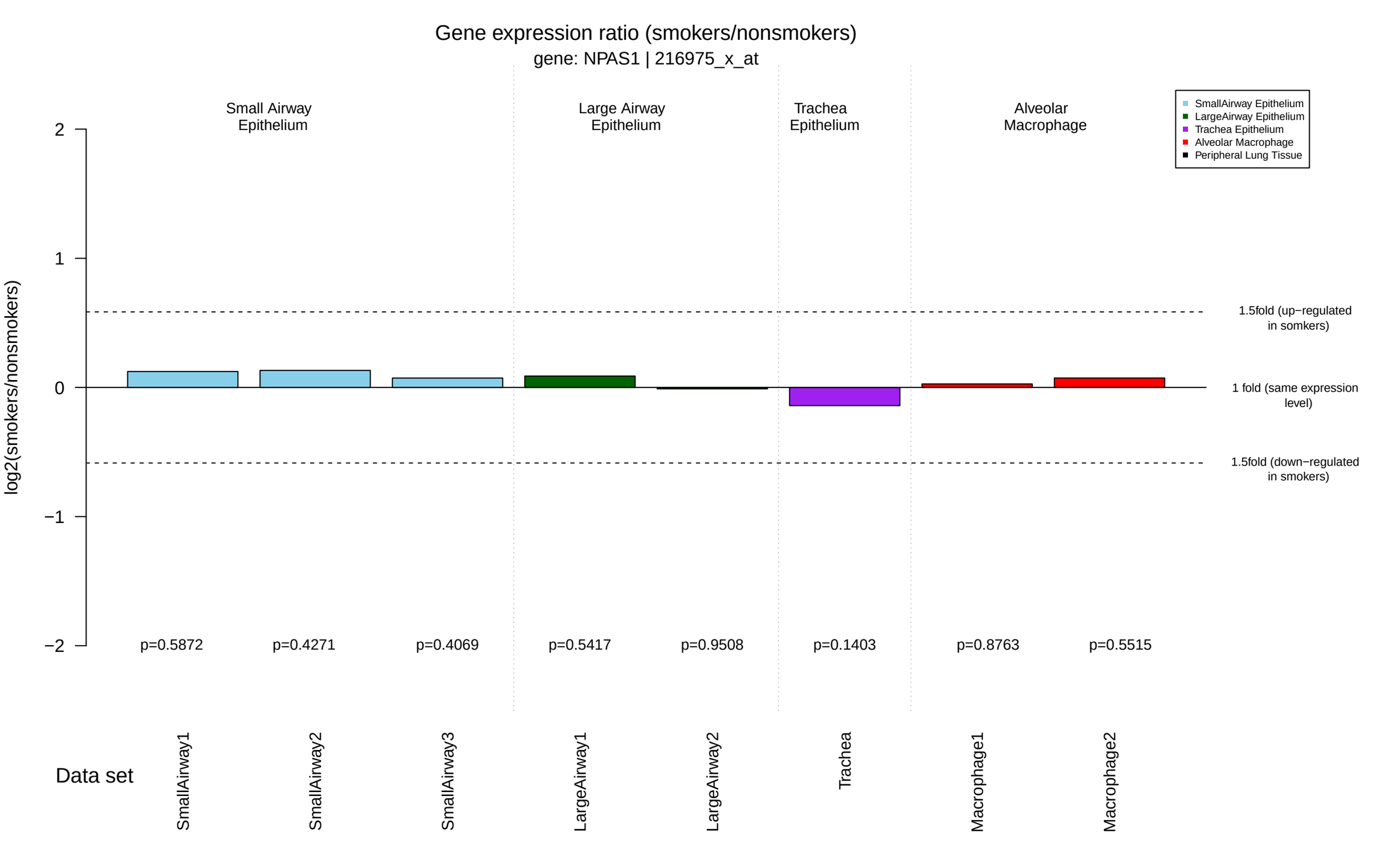
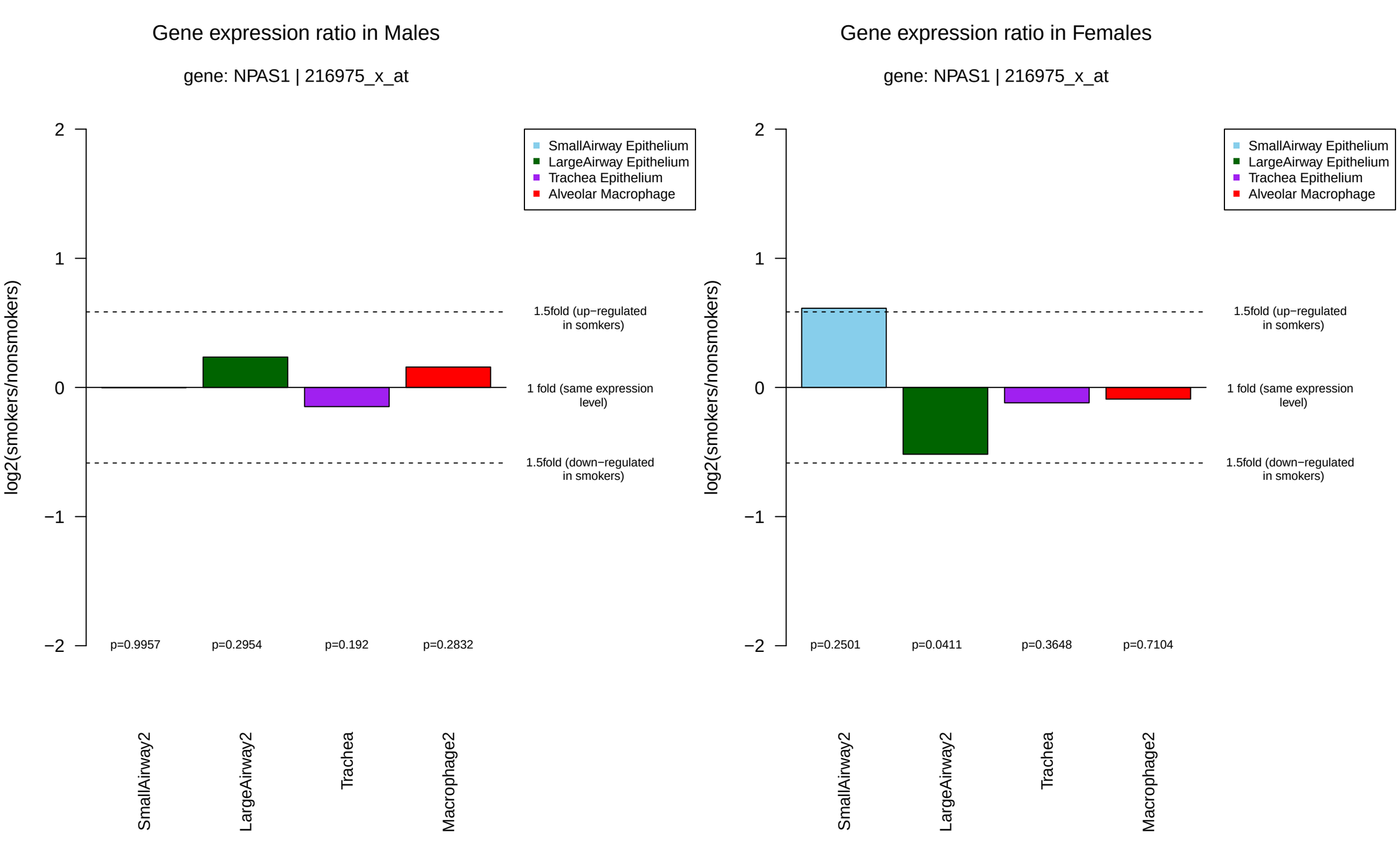
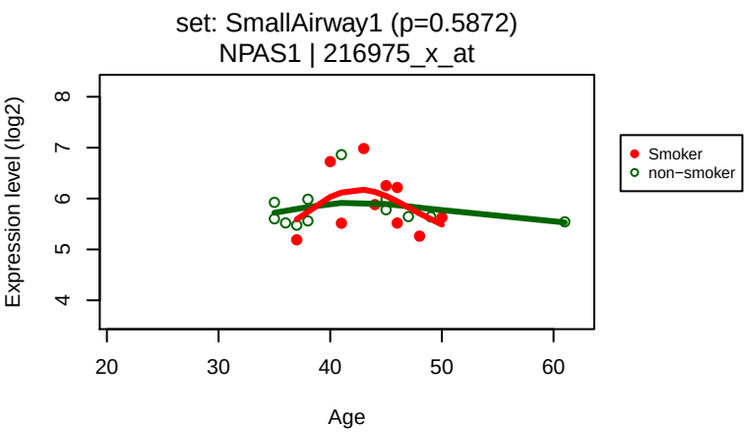
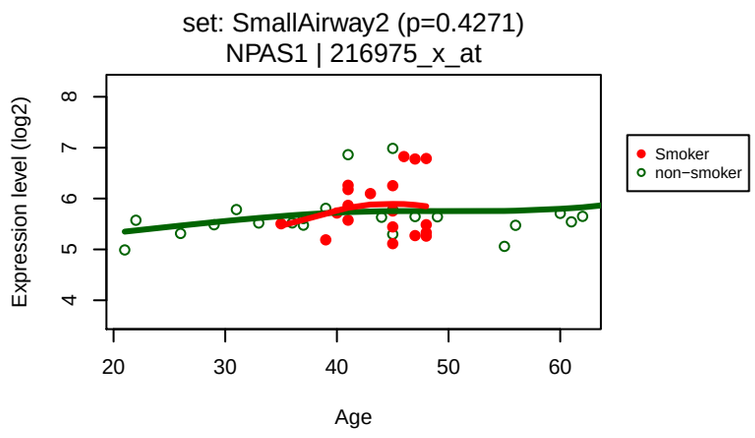
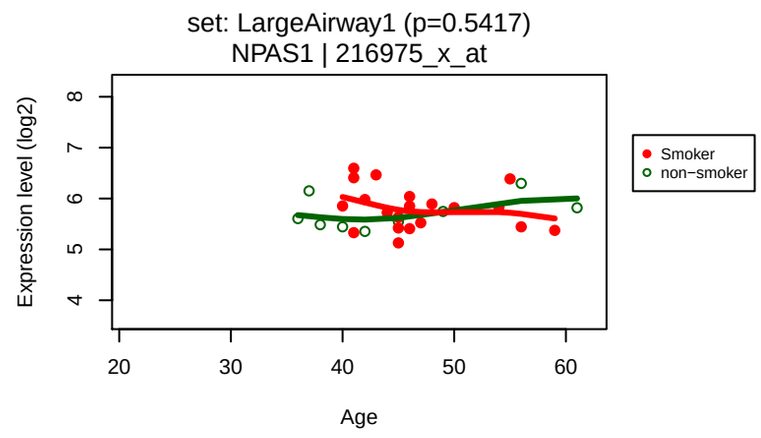

Gene expression of smokers and nonsmokers
gene: NPAS1 | 216975_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5872
p=0.4271
p=0.4069
p=0.5417
p=0.9508
p=0.1403
p=0.8763
p=0.5515
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

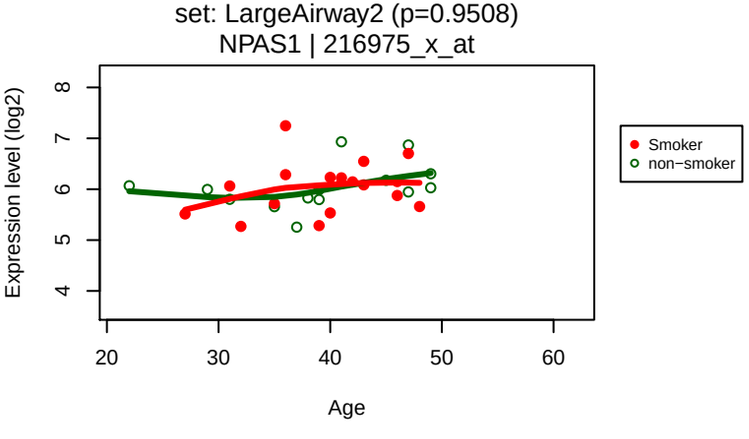

Gene expression ratio (smokers/nonsmokers)
gene: NPAS1 | 216975_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5872
p=0.4271
p=0.4069
p=0.5417
p=0.9508
p=0.1403
p=0.8763
p=0.5515
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

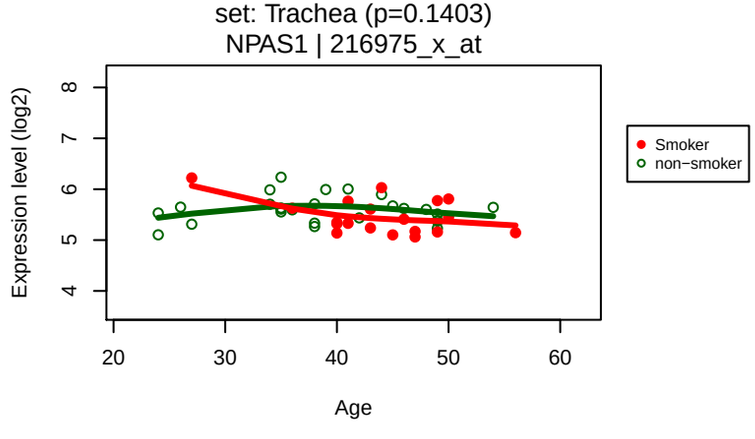

Gene expression ratio in Males
gene: NPAS1 | 216975_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9957
p=0.2954
p=0.192
p=0.2832
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPAS1 | 216975_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2501
p=0.0411
p=0.3648
p=0.7104
SmallAirway2
LargeAirway2
Trachea
Macrophage2

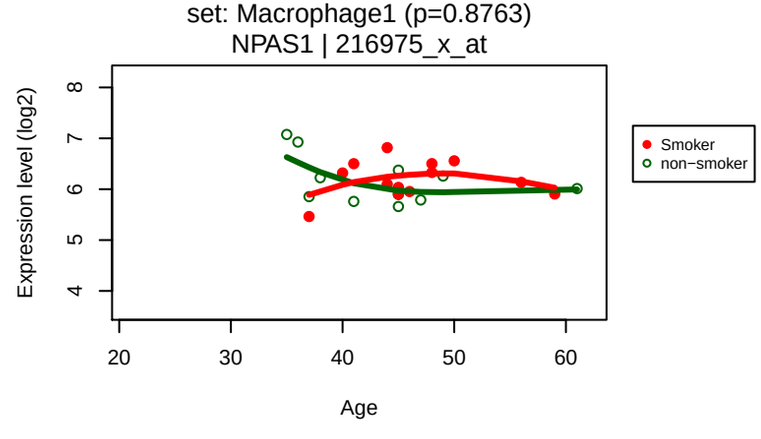

set: SmallAirway1 (p=0.5872)
NPAS1 | 216975_x_at
Expression level (log2)
Age
Smoker
non−smoker

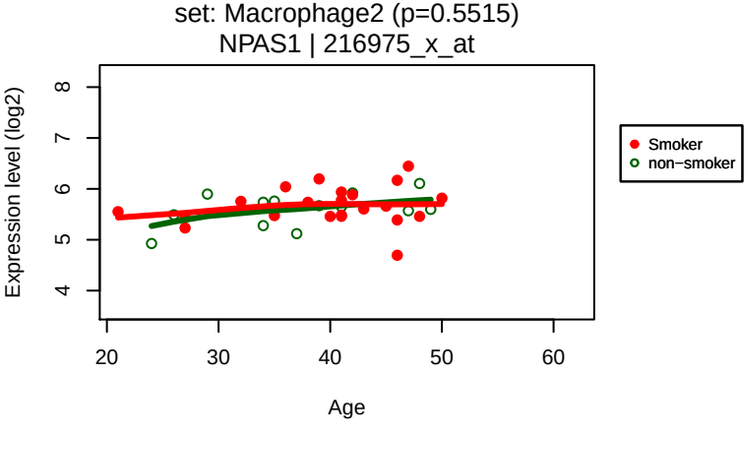

set: SmallAirway2 (p=0.4271)
NPAS1 | 216975_x_at
Expression level (log2)
Age
Smoker
non−smoker

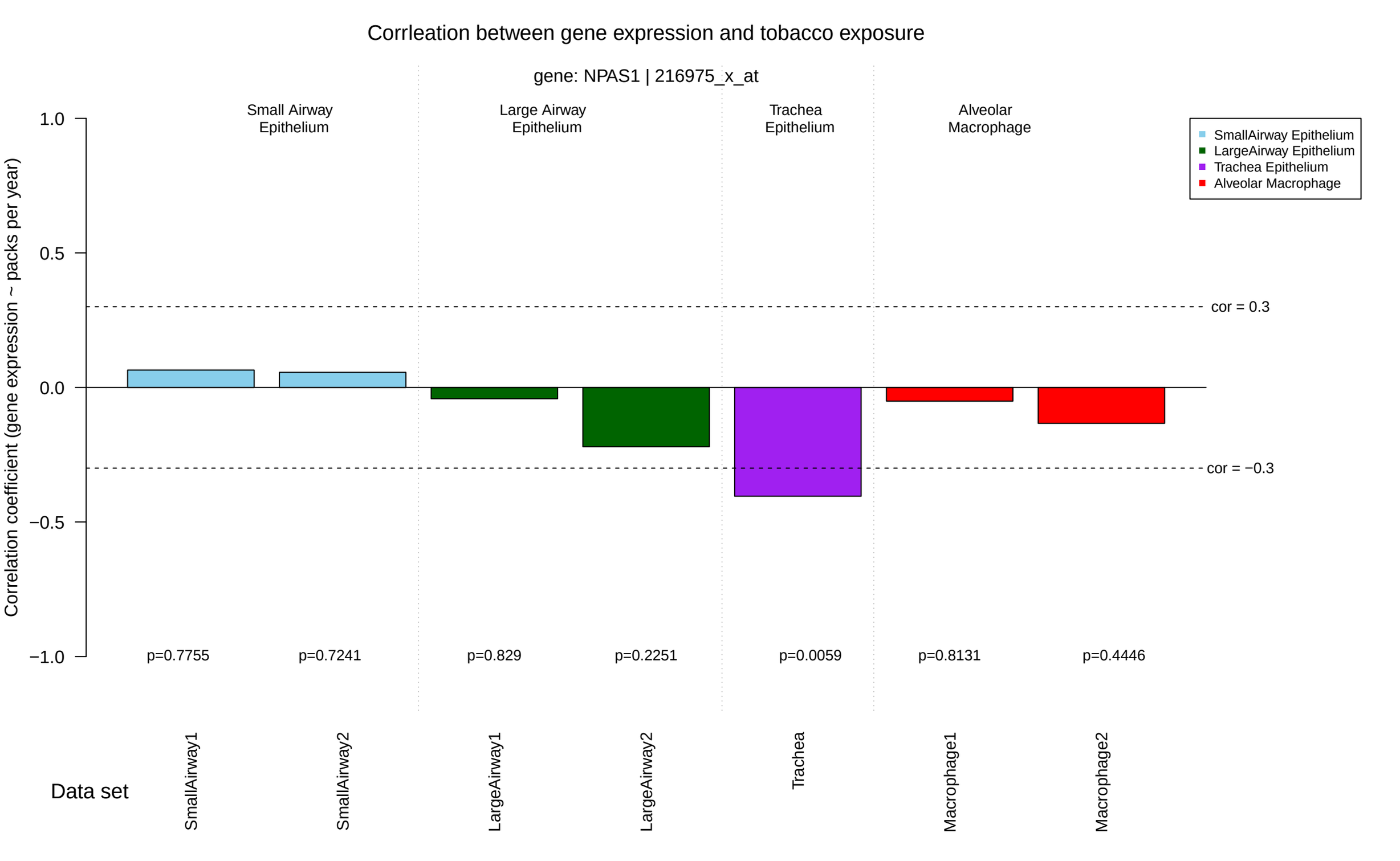

set: LargeAirway1 (p=0.5417)
NPAS1 | 216975_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9508)
NPAS1 | 216975_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1403)
NPAS1 | 216975_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8763)
NPAS1 | 216975_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5515)
NPAS1 | 216975_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPAS1 | 216975_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7755
p=0.7241
p=0.829
p=0.2251
p=0.0059
p=0.8131
p=0.4446
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2